IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-404

1. Topic (12 words max)

SignSight: Empowering Accessibility with Tamil Sign Language Recognition Using Machine Learning

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

The use of Tamil Sign Language (TSL) together with new technologies like Artificial Intelligence (AI), Machine Learning (ML), and Computer Vision can help solve many communication problems faced by the hearing-impaired community in Sri Lanka. TSL is the main way of communication for them, but it does not have strong rules or standards. Because of this, it is often hard to share complete thoughts and ideas. Current translation tools are also limited because they cannot properly handle real-time signs, emotions, or meanings in context, making them less useful in daily life.[1][2]

This research focuses on four main parts. The first part is building an audio/video-to-TSL conversion system, which can translate Tamil speech and videos into sign language. The second part is creating a dynamic sign detection system that can recognize moving gestures and continuous signing. The third part is developing a TSL learning platform where people can study signs, take quizzes, and use text-to-sign translation to improve their skills. The fourth part is about emotion and context-aware TSL tools for children, especially those with cochlear implants. This will help children learn both signs and emotional expressions, making communication more natural.[2][3]

Modern AI and ML methods like CNNs, RNNs, and LSTMs can be used to make these systems work better.[5] They can improve accuracy in real-time recognition, context-based translation, and expressive sign communication. However, it is important to make sure these systems are reliable, because mistakes could cause confusion and more communication problems.[2][4]

Another challenge is handling Tamil audio correctly, since people speak with different accents, pronunciations, and background noise. Good preprocessing methods are needed to clean and prepare speech before translation. For children, training exercises should be personalized to track their progress and give feedback to improve their learning.[3]

Finally, these systems should be built to be low-cost, accurate, and scalable, so that more people can use them. If done successfully, AI-based TSL systems can create better inclusivity by allowing hearing-impaired people to take part more fully in education, work, and society. [2][3]

References:

[1] Deep Kothadiya, Chintan Bhatt, Krenil Sapariya, Kevin Patel, Ana-Belén, Gil-Gonzálezand and Juan M. Corchado, "Deep Sign: Sign Language Detection and Recognition Using Deep Learning Electronics", 2022.

[2] Rathnayake R.K.D.M.P, Wijekoon W.M.S.T, Rajapakse K.G, Rasanjalee K.A. and Dilshan De Silva, "Real-Time Sign Language Translator", International Journal of Engineering and Management Research, December 2022.

[3] Pumudu Fernando and Dr. Prasad Wimalaratne, GSTF Journal on Computing (JOC), "Sign Language Translation Approach to Sinhalese Language", Issue 1; 2016.

[4] H.H.S.N. Haputhanthri, H.M.N. Tennakoon, M.A.S.M. Wijesekara, B.H.R. Pushpananda,
H.N.D. Thilini, "Multi-modal Deep Learning Approach to Improve Sentence level Sinhala Sign Language Recognition", International Journal on Advances in ICT for Emerging Regions 2023.

[5] P. Rishi Sanmitra, V. V. Sai Sowmya and K. Lalithanjana, "Machine Learning Based Real Time Sign Language Detection", International Journal of Research in Engineering, Science and Management Volume 4, Issue 6, June 2021.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

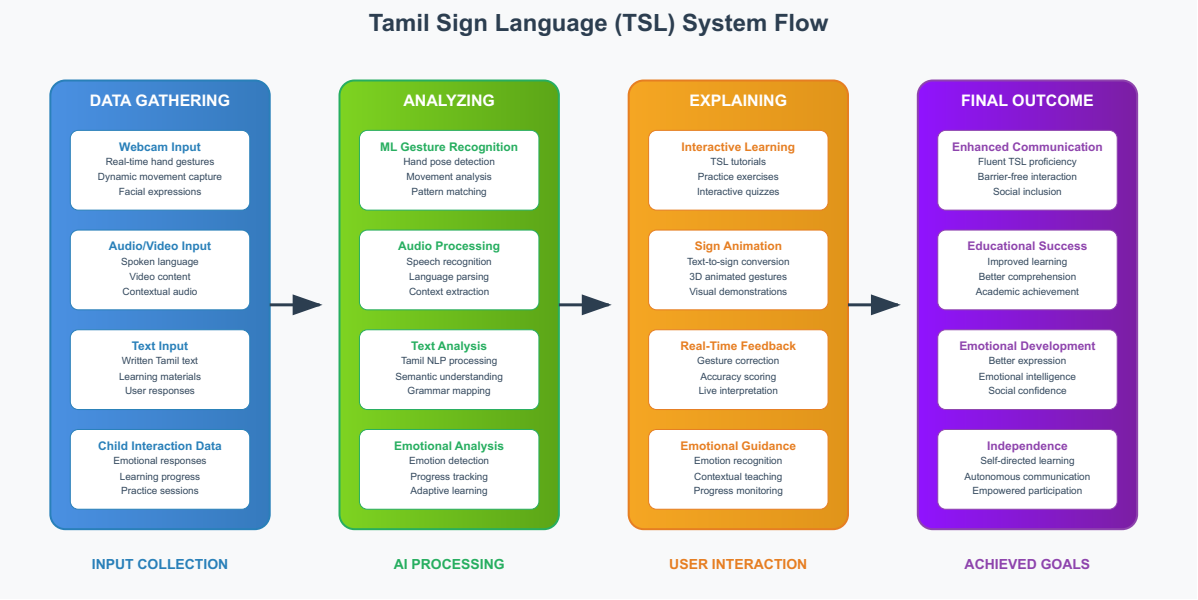
The purpose of the Tamil Sign Language (TSL) system is to help children with Standard and Emotional Conversations and to enhance communication for deaf people. It consists of four primary parts:

1. Tamil Sign Language Learning Module:
   • This module provides foundational training in TSL, enabling users to learn and practice signs. It includes interactive quizzes and a tool that converts written text into sign language for better understanding.

2. Real-Time Sign Gesture Recognition:
   • Through webcam input, this system detects and interprets dynamic hand gestures on the spot, using machine learning algorithms to accurately recognize sign language movements.

3. Audio/Video to TSL Conversion System:
   • Converts spoken audio or video content into animated Tamil sign language, helping users who depend on sign language to comprehend verbal communication.

4. Emotionally and Contextually Aware TSL Support for Children:
   • Offers tailored exercises that teach children to identify and convey emotions using TSL, while monitoring their progress to enhance their emotional expression and communication skills.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

This research aims to develop a comprehensive Tamil Sign Language (TSL) system for the hearing-impaired community in Sri Lanka, requiring interdisciplinary expertise in machine learning, computer vision, speech processing, and sign language linguistics.

Specialized Domain Expertise: -

1. **Machine Learning and Computer Vision:**
   - Expertise in Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Random Forest (RF) for gesture recognition and dynamic sign detection.
   - Skills in using frameworks like TensorFlow and OpenCV for real-time dynamic sign processing.
2. **Audio and Speech Processing:**
   - Proficiency in speech recognition to convert Tamil audio to text and map it to TSL signs.
   - Knowledge in tokenization and phoneme analysis for accurate conversion of spoken Sinhala into visual signs.
3. **Sign Language Linguistics:**
   - Understanding Tamil grammar and TSL-specific linguistic challenges for accurate translation of written Tamil into sign language.
   - Expertise in incorporating prefixes and suffixes into TSL signs for proper linguistic structure.
4. **Software Development:**
   - Expertise in web-based application development and API integration for real-time interaction with machine learning models.

Knowledge:-

1. Machine Learning and AI:
   - Knowledge of RNNs and LSTMs for gesture recognition and audio-to-sign conversion.
2. Sign Language Linguistics:
   - Knowledge of Sinhala grammar and its translation challenges into SSL.
3. Speech Recognition:
   - Understanding of speech-to-text systems and phonetic analysis for accurate conversion.
4. Software Architecture:
   - Knowledge of designing scalable systems and integrating APIs for seamless system operation.

Data Requirements: -

1. **SSL Gesture Databases:** Datasets of static and dynamic SSL gestures for model training.
2. **Audio-Visual Datasets:** Sinhala speech and SSL signs for training the audio-to-sign system.
3. **Participant Data:** Data from cochlear-implanted children to support the vocal training module.

8. Objectives and Novelty

Main Objectives

1. **Develop a comprehensive learning platform for Tamil Sign Language (TSL):**
   - Create an interactive system for teaching TSL with structured lessons, assessments, and quizzes.
   - Enable users to learn both static and dynamic signs, improving communication for the hearing-impaired community in Sri Lanka.

2. **Enhance dynamic sign recognition using machine learning:**
   - Build a system capable of recognizing and interpreting dynamic TSL gestures using advanced computer vision techniques.
   - Implement real-time recognition of gestures, improving accessibility and communication for TSL users.

3. **Bridge the gap between spoken Tamil and TSL through audio/video conversion:**
   - Develop a tool that converts Tamil audio and video inputs into corresponding TSL signs.
   - Provide an intuitive solution that helps hearing-impaired individuals understand spoken language via visual representation.

4. **Develop emotion and context-aware TSL enhancement for children:**
   - Design interactive exercises to help children, including those with cochlear implants, recognize and express emotions through TSL.
   - Track progress and provide real-time feedback to improve expressive and communication skills in children.

5. **Foster inclusivity and accessibility for the hearing-impaired community in Sri Lanka:**
   - Create a user-friendly platform that enhances communication, education, and interaction opportunities for the hearing-impaired.
   - Ensure that technology promotes social inclusion, making daily life and learning easier for TSL users.

Member Name with Registration No	Sub Objective	Tasks	Novelty
Farsith M F M IT22354556	Develop a system for converting Tamil Audio/Video to Tamil Sign Language (SSL)	1. Build a speech recognition module to convert Tamil audio to text using machine learning. 2. Train logistic regression models to predict Tamil Sign Language (TSL) gestures from audio/video inputs. 3. Develop a user-friendly API for converting multimedia inputs into Tamil Sign Language (TSL) gestures.	1. Tokenize and map Tamil text to legacy character representations for TSL compatibility. 2. Introduce scalable and low-resource models to map auditory, visual inputs to visual signs. 3. Real-time conversion of audio and video content into visual TSL improves accessibility.

Mark J J IT22328120	Create a dynamic sign detection system using computer vision	1. Develop a dataset of dynamic TSL gestures captured with a webcam using OpenCV and Mediapipe. 2. Train the system to interpret transitional movements in dynamic signs. 3. Implement a backend API for real-time gesture recognition and integration into the platform.	1. Train LSTM-based neural networks on motion sequences to recognize gestures with high precision. 2. Introduce preprocessing techniques like key point extraction and encoding for reliable sign recognition. 3. Real-time video processing ensures seamless user experience, catering to real-world scenarios.
Nethmi G K IT22546166	Develop a learning system for Tamil Sign Language (SSL)	1. Design interactive lessons for TSL, including static signs and multiple-choice assessments (MCQs). 2. Implement a text-to-sign language interpreter for written Tamil text. 3. Integrate an evaluation system to track learning progress through MCQs and quizzes.	1. Introduce a structured three-level learning platform incorporating Tamil signs for letters, words, and sentences, ensuring progressive learning experiences. 2. Utilize a database- driven approach with prefix/suffix additions to TSL signs to address grammar complexities. 3. Gamified assessments tailored to evaluate comprehension at each learning stage.

Sandamini G H D H IT22291882	Develop an Emotionally and Contextually Aware TSL Support for Children:	1. Use CV + audio models to detect emotions (facial expressions, tone). 2. Apply NLP to capture contextual meaning in Tamil sentences. 3. Enhance TSL output with facial cues + contextually accurate signs via API.	1. Combine face and voice emotion detection to make TSL signs show feelings and match the emotional tone automatically. 2. Use smart Tamil language processing to understand context and create age-appropriate TSL signs for children's learning needs. 3. Build adaptive emotion-based learning that changes teaching style based on how children feel and respond during TSL practice sessions.

9. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Dr	Prasanna	Sumathipala	
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	

b) Does the proposed topic exhibit novelty?

Yes		No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

IT4010 Research Project – 2025 July Topic Assessment Form

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.